EEE463 Wireless Communication Project Report

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***Abstract*— The architecture of harmonious wireless networks is examined in this study from the standpoint of interference control. When helpful and interfering signals coexist, they create a complementary relationship that increases the network's overall capacity to transmit data. At the network level, these signals exhibit a symbiotic structure, despite playing opposing roles within individual communication links. The Chinese concept of balanced opposition, symbolized by the "yin" and "yang" philosophy, is reflected in this duality. Interfering and helpful signals must coexist in harmony for a wireless network to function at its best. Interference management is crucial in this situation, and in order to optimize network performance, sophisticated, integrated techniques must be created.**

Key words: duality, complementarity principle, yin and yang, interference management, harmonious wireless networks

I. Introduction

Orthogonal multiple access (OMA) techniques have been used in conventional 2G, 3G, and 4G systems to avoid user interference. By assigning unique frequency resources to every user, these methods seek to eliminate signal collisions. In particular, single-carrier frequency-division multiple access (SC-FDMA) is favored for uplink transmission because of its lower peak-to-average power ratio (PAPR), whereas orthogonal frequency-division multiple access (OFDMA) is used for downlink transmission in 4G systems. By preventing subcarrier overlap, these structures remove user-to-user direct interference.

Non-orthogonal multiple access (NOMA) techniques have gained prominence as a result of the introduction of 5G systems, which have made higher spectral efficiency a top priority. For example, 5G's sparse code multiple access (SCMA) technique permits user overlap while still allocating orthogonal subcarriers. This method improves system performance overall, even though there is some interference.

The overall data capacity of the system is also increased by heterogeneous network architectures, which are made up of densely deployed femto and pico cells inside the coverage area of conventional macro cells. Performance improvements are still possible in such structures, despite the fact that cross-tier and inter-cell interference is unavoidable. Furthermore, despite mutual interference between the two systems, hybrid networks—where LTE-U and Wi-Fi systems coexist—offer the possibility of increasing spectral efficiency in 5G and beyond.

The crucial question at this stage is not whether interference is present in the network, but rather whether it can be separated from signals that are helpful. Advanced interference management techniques can be used to suppress, mitigate, or even exploit interference if it can be distinguished. This viewpoint is consistent with the idea of "duality," where interference and helpful signals coexist in a mutually beneficial and complementary manner.

INTERFERANCE-SIGNAL DUALITY IN WIRELESS NETWORKS

At this point, whether or not interference in the network can be separated from helpful signals is more important than whether it exists at all. Advanced interference management techniques can suppress, reduce, or even use the interference to the system's advantage if it can be identified. The idea of duality, in which helpful signals and interference coexist in a complementary and mutually beneficial relationship, can be used to explain this viewpoint.

One user's desired signal could be another's interference. For example, a signal meant for your phone might cause your neighbor's phone to make noise. In other words, depending on the situation, a signal may be advantageous or detrimental. The key is to identify and distinguish it. We should learn to coexist with such signals rather than attempting to eradicate them..

Signals and noise are not the only factors that wireless networks use to function. Like Yin and Yang, the two come together to form a whole. One is useless without the other, much like night and day. They ought to be viewed as complementary elements as a result. The concept of Yin-Yang, in which the black side (Yin) represents the interfering signal and the white side (Yang) represents the useful signal, is a good example of this idea in Chinese philosophy.

Figure1 Complementarity principle of wireless networks using “yin” and “yang”.

STRATEGIES FOR INTERFERENCE MANAGEMENT

In wireless communication systems, interference is managed using five basic techniques. By directing transmission nulls in the direction of interference, nulling seeks to remove it. In order to confine interference to a smaller dimensional space and free up resources for desired signals, alignment aims to align several interfering signals into the same subspace. Designing signal paths so that conflicting signals cancel each other out at the receiver is known as neutralization. Using methods like filtering, coding, or power control to lessen the effects of interference is known as mitigation. The most sophisticated strategy is exploitation, which uses cooperative communication or network coding, for instance, to improve system performance rather than just suppressing interference. These methods demonstrate a change in viewpoint from viewing interference as a completely negative factor to appreciating and harnessing its potential.

II. EXPERIMENTS AND RESULTS

Nulling → To totally remove the interference

Alignment → The process of aligning conflicting signals in the same signal dimension or space

Neutralization → To enable conflicting signals to eliminate one another

Mitigation → To lessen the effects of interference

Exploitation → Making use of interference for the system's advantage

2.1) **Interference Nulling** :

A null space, where the useful signal has zero sensitivity, is the target of the interference. In other words, it is flung in a direction that the signal cannot see, rendering it useless there.

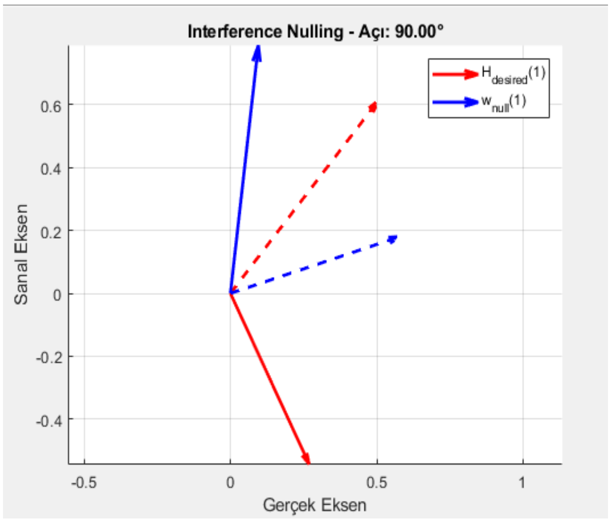

Figure2: Interference Nulling — Visualization of desired signal and interference beamforming vectors in the complex plane.

The efficiency of interference nulling in a 2x2 MIMO system is illustrated in this figure. Whereas the blue vector displays the beamforming direction of the interfering signal (w_{null}), the red vector indicates the desired signal direction at the receiver (H_{desired}).

The interference is directed into the desired signal's null space, as indicated by the automatic adjustment of the angle between these two vectors to 90.00°.

This condition can be expressed mathematically as H_{desired}^H * w_{null} = 0, which indicates that the interference is orthogonal to the desired signal and, as a result, has no effect at the receiver.

The main objective of interference nulling is to direct the interference in a spatial direction that the intended receiver cannot detect, and this orthogonality guarantees that the receiver is totally unaffected by the interference.In order to reduce interference without sacrificing the intended signal quality, interference nulling is therefore essential in multi-antenna systems (MIMO).

2.2) **Interferance Aligment :**

In multi-user wireless systems, interference alignment (IA) is a technique that reduces the dimension occupied by interference by confining multiple interfering signals into the same subspace.

This increases the degrees of freedom (DoF) and boosts system capacity by making more space for desired signals.

In multiple-input multiple-output (MIMO) systems, beamforming is frequently used to implement interference alignment.

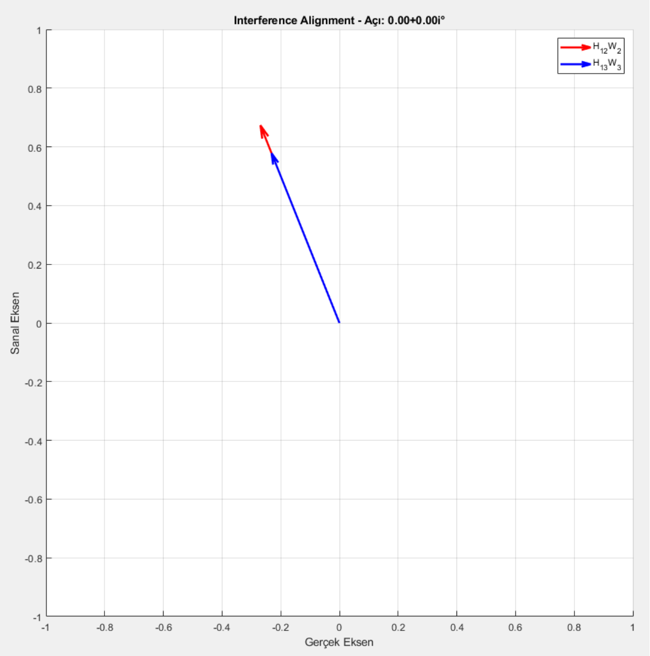

Figure3: Interference Alignment – Directional Overlap of Interfering Signals at a Common Receiver

By aligning the interference from two transmitters (Tx2 and Tx3) at a single receiver (Rx1), this experiment illustrates the idea of interference alignment (IA).

The appropriate beamforming vectors (W_2, W_3) were calculated in such a way that the vectors $H_{12}W_2$ (red) and $H_{13}W_3$ (blue), which represent the two interfering signals, point in the same spatial direction.The interference signals were successfully aligned when the angle between the two vectors was determined to be 0.00°.By reducing the interference dimensionality, this alignment increases the system's total degrees of freedom by providing more room for the intended signal.

2.3) **Interference Neutralization :**

Interference Neutralization is a technique that uses beamforming to properly adjust the amplitude and phase of interfering signals from various sources so that they cancel each other out at the receiver.Throughput and system reliability are increased by making sure that these signals arrive in opposite directions, which reduces net interference.

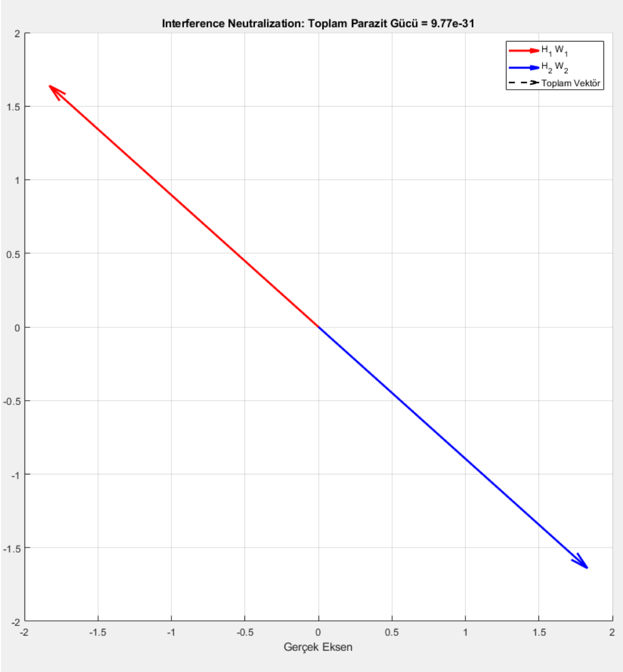

Figure4: Interference Neutralization — Vectorial Cancellation of Two Interfering Signals

Interfering signals from two separate transmitters (Tx1 and Tx2) are set up to cancel each other out at a common receiver (Rx) in this interference neutralization scenario, as shown in the figure.

The sum of the oppositely oriented vectors $H_1 \cdot W_1$ (red) and $H_2 \cdot W_2$ (blue) is almost zero.

As a result, the interfering signals' detrimental effect on the intended transmission is successfully removed, and the receiver experiences no interference.

2.4) **Interference Mitigation :**

In wireless communication systems with multiple users, interference mitigation is a class of techniques used to lessen or suppress the impact of interfering signals at the receiver.

The goal is to maximize the desired signal's reception while reducing the impact of interference, usually by using adaptive receive beamforming or spatial filtering.

To increase the Signal-to-Interference-plus-Noise Ratio (SINR) and overall system performance, methods like MMSE, Zero-Forcing (ZF), and SINR-based optimization are frequently employed.

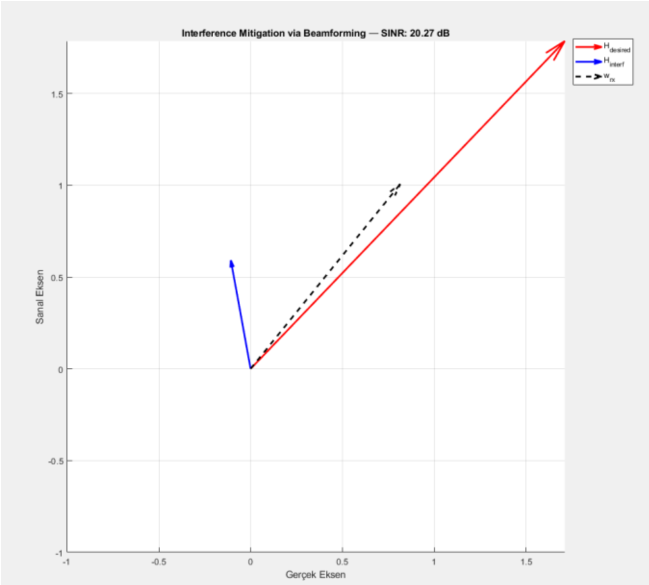

Figure5: Interference Mitigation — Beamforming Maximizing SINR

This figure illustrates a more sophisticated method of interference mitigation where adaptive receive beamforming is used to tolerate and control interference rather than completely suppress it.

The complex plane shows the interference vector H_{interf} (blue) and the desired signal direction H_{desired} (red).

By strategically aligning with the desired signal while accepting interference from an angularly distinct direction, the beamforming vector w_{rx} (black dashed) is optimized to maximize the signal-to-interference-plus-noise ratio (SINR)`. This is not accomplished by nullifying the interference.

In contemporary wireless communication, where interference can occasionally be constructively exploited or tolerated to preserve degrees of freedom and improve overall capacity, this approach reflects a pragmatic and forward-looking approach.The resulting SINR of 20.27 dB illustrates the success of this adaptive mitigation, enabling high-quality reception even in the presence of interference.

2.5) **Interference Exploitation :**

In conventional wireless communication systems, interference is typically regarded as a harmful factor that must be suppressed or eliminated. However, in modern multi-antenna architectures, when the characteristics and direction of the interference are known, it can be repurposed as a constructive element. This technique, known as interference exploitation, aligns the interference vector in phase with the desired signal, thereby boosting the overall received signal strength. As a result, the combined signal lies farther from the decision boundaries, improving symbol detection accuracy. This approach not only reduces the error probability but also enhances spectral efficiency, particularly in densely populated communication environments.

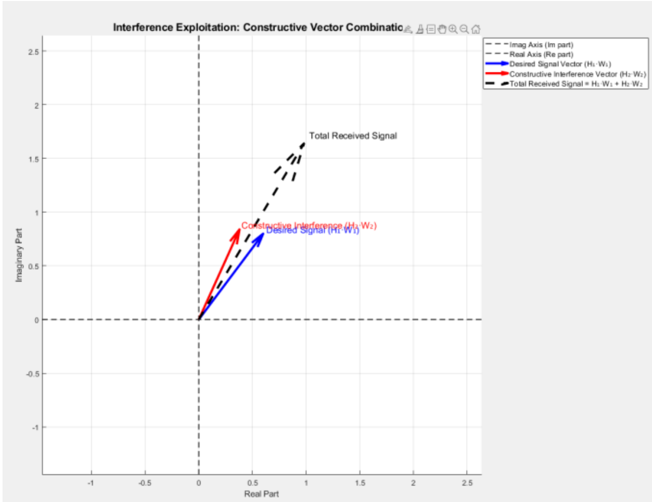

Figure6: Interference Exploitation

The desired signal ($H_1 \cdot W_1$, blue) and the known interference ($H_2 \cdot W_2$, red) are aligned in phase to constructively combine.The resulting signal (dashed black) lies further from decision boundaries, illustrating how interference can be leveraged as a constructive resource rather than a harmful disturbance.

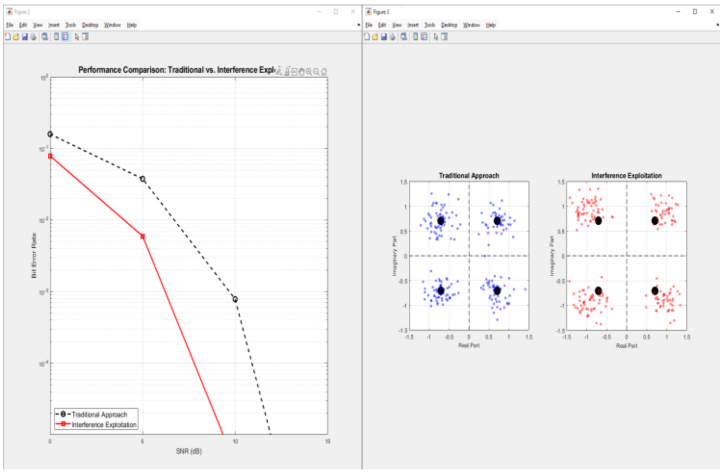

Figure7:Constructive Use of Interference in Wireless Communication

Figure 7. The figure illustrates the system performance and symbol distribution achieved through the *Interference Exploitation* technique. The graph on the left demonstrates that treating interference as a constructive component—rather than suppressing it—leads to significantly lower bit error rates compared to traditional approaches. On the right, constellation diagrams provide an intuitive interpretation of this gain: while traditional methods result in symbols clustered near decision boundaries with higher uncertainty, interference exploitation ensures that

symbols are pushed into more stable and reliable regions. This is accomplished by combining known and controllable interference signals with the desired signal in a phase-aligned manner, thereby enhancing signal strength. Particularly beneficial in dense networks, MIMO systems, and satellite communications, this strategy transforms interference from a harmful disruption into a **resource that constructively contributes to signal quality**, marking a paradigm shift in how interference is approached in wireless communication systems.

III. Performance Comparison

In modern wireless communication systems, interference remains one of the primary bottlenecks limiting overall system efficiency. As a result, several techniques have been developed to manage interference, each offering distinct advantages and trade-offs both theoretically and practically. The five core strategies are summarized below:

Nulling: Aims to completely eliminate interference by steering nulls in the direction of interfering signals at either the transmitter or receiver. While effective, this can also reduce the power of the desired signal and requires precise directional control.

Alignment: Aligns multiple interfering signals into the same subspace, making it easier for the receiver to separate the desired signal. This method is especially powerful in multi-user MIMO systems, but relies heavily on accurate channel knowledge and phase coordination.

Neutralization: Uses the destructive combination of multiple interfering signals so that they cancel each other out at the receiver. While elegant in theory, this approach typically requires coordinated multi-transmitter systems and strict synchronization.

Mitigation: Suppresses interference at the receiver through beamforming or filtering. It offers flexibility but often demands additional hardware complexity and computational effort.

Exploitation (This Work): Interference is not suppressed — it is embraced and redirected for system benefit.By aligning the phase and direction of known interference with the desired signal, the total received signal is constructively enhanced. This improves power utilization, increases the distance from decision boundaries, and results in more reliable symbol detection.

As shown in Figure 7, especially in high-SNR regimes, interference exploitation consistently achieves lower bit error rates compared to conventional suppression-based strategies. The constellation diagrams visually reinforce this outcome: while other techniques leave symbols vulnerable near decision boundaries, the exploitation approach pushes them deeper into safer zones, improving detection certainty.

While other methods seek to silence interference, interference exploitation chooses to listen and redirect it.
This leads not only to performance gains, but also to increased agility and adaptability in wireless systems.

In summary, interference exploitation stands out not only for its technical merits, but also as a paradigm shift in how interference is perceived and utilized in next-generation communication frameworks.

IV. Summary and Concluding Remarks

In this study, we examined a range of interference management techniques—namely nulling, alignment, neutralization, mitigation, and exploitation—each aiming to improve communication performance under interference-limited conditions. Through both theoretical discussion and visual simulations, we explored their core principles and practical implications in multi-user wireless systems.

Our MATLAB-based visualizations illustrated the geometric behavior of each method:

In nulling, the interference vector was fully canceled by beamforming in the orthogonal direction.

In alignment, interference components from multiple users were steered into the same subspace to preserve spatial degrees of freedom.

Neutralization showed that interference from two sources could destructively cancel each other at the receiver.

In mitigation, beamforming was applied at the receiver to suppress interference and maximize SINR.

Most notably, our interference exploitation simulations demonstrated that, when properly aligned, interference can be constructively combined with the desired signal—resulting in stronger overall reception and reduced error rates.

Performance graphs clearly revealed that interference exploitation consistently outperforms traditional suppression-based techniques, particularly in moderate-to-high SNR regions. The constellation diagrams further supported this finding, showing how constructive interference pushes received symbols deeper into decision-safe zones, reducing the likelihood of symbol errors.

Rather than viewing interference as a threat, this work presents it as a potential ally—a resource to be directed rather than rejected.

In conclusion, while all five techniques offer valuable contributions to the challenge of interference, interference exploitation stands out as a more adaptive and forward-looking strategy. It not only improves performance metrics like SER and BER but also encourages a paradigm shift in how wireless systems interact with interference. As networks evolve into more complex, dense, and cooperative architectures, embracing and harnessing interference may no longer be a novelty—but a necessity.

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References

[1] **Li, A., Spano, D., Krivochiza, J., Domouchtsidis, S., Tsinos, C. G., Masouros, C., Chatzinotas, S., Li, Y., Vucetic, B., & Ottersten, B.** *Interference Exploitation via Symbol-Level Precoding: Overview, State-of-the-Art and Future Directions* arXiv preprint arXiv:1907.05530, 2019.

[2] **Trabelsi, N., Chaari Fourati, L., & Chen, C. S.** *Interference Management in 5G and Beyond Networks* arXiv preprint arXiv:2401.01608, 2024.

[3] Costa, M. Writing on Dirty Paper IEEE Transactions on Information Theory, vol. 29, no. 3, pp. 439–441, 1983.

[4] Masouros, C., & Zheng, G.

Exploiting Constructive Interference in Wireless Communications: A Survey IEEE Communications Surveys & Tutorials, vol. 17, no. 2, pp. 1290–1321, 2015.

[5**]** **Li, A., Masouros, C., & Ottersten, B.** *Symbol-Level Precoding for Multiuser MISO Systems with Constructive Interference*
IEEE Transactions on Signal Processing, vol. 64, no. 5, pp. 1328–1342, 2016.

[6] W. Liu, "Harmonious Wireless Networks: Perspective of Interference Management," *Intelligent and Converged Networks*, vol. 2, no. 3, pp. 198–204, Sep. 2021, doi: 10.23919/ICN.2021.0013.

[7] A. Celik, A. Chaaban, B. Shihada, and M. S. Alouini, Topology optimization for 6G networks: A network information-theoretic approach, *IEEE Veh. Technol. Mag.*, vol. 15, no. 4, pp. 83–92, 2020.
[5]
B. Soret, A. De Domenico, S. Bazzi,

[8] Q. Wang, R. Zhang, L. L. Yang, and L. Hanzo, Nonorthogonal
multiple access: A unified perspective, *IEEE Wirel. Commun.*, vol. 25, no. 2, pp. 10–16, 2018.

[9] M. Ali, S. Qaisar, M. Naeem, W. Ejaz, and N. Kvedaraite,
LTE-U WiFi HetNets: Enabling spectrum sharing for 5G/Beyond 5G systems, *IEEE Internet Things Mag.*, vol. 3, no. 4, pp. 60–65, 2020.

[10] E. Hossain, M. Rasti, H. Tabassum, and A. Abdelnasser,
Evolution toward 5G multi-tier cellular wireless networks: An interference management perspective, *IEEE Wirel. Commun.*, vol. 21, no. 3, pp. 118–127, 2014.

[11] N. Lee and R. W. Jr. Heath, Advanced interference management technique: Potentials and limitations, *IEEE Wirel. Commun.*, vol. 23, no. 3, pp. 30–38, 2016.

[12] Wikipedia, Yin and yang, https://en.wikipedia.org/wiki/Yin_and_yang, 2021.